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20240045

Architecture

All modern computing rely on the foundation of computer architecture concepts, CPUs and the Von Neumann architecture. Together they re-imagine how computers process, retrieve, and store instructions. The Von Neumann Architecture was proposed and authored by mathematician John Neumann in 1945. It describes a system where both programs instructions and data share the same memory space, connected to the CPU via Bus systems. The design is the template for virtually all modern computers.

The CPU executes programs through the Fetch-Decode-Execute (FDE) cycle. First, in the fetch stage CPU reads the next instruction from the RAM using the program counter (PC) and memory address register (MAR). Next, the decode stage, the control unit (CU) interprets the instruction's opcode and prepares the necessary resources. Finally! In the execute stage, the arithmetic logic unit (ALU) carries out arithmetic and logical operations like: addition, comparison, and data transfer. The end results are stored.

Understanding this cycle is critical in IT because it underlines the design for modern processor computing, operating system scheduling, compiler organization, and performance tuning. Every program, from a web browser to a large-scale server, ultimately running as a sequence of these three steps repeatedly millions of times per second.

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The purpose of this diagram is to illustrate the Von Neumann architecture. How modern computers process and store program instructions and data